

Bank Loan Risk & Approval Analysis

End-to-End Data Analytics Portfolio Case Study

1. PROJECT OVERVIEW & BUSINESS VALUE

This project presents an end-to-end data analytics workflow implemented on a retail bank credit portfolio containing **2,500 applicant records**. The analysis addresses key risk parameters, identifies critical loan approval drivers, evaluates debt-to-income limits, and surfaces clear demographic segments to optimize bank credit risk models.

SQL (PostgreSQL)

Python (Pandas & NumPy)

Power BI Desktop

DAX & Power Query

2. ADVANCED DATABASE ANALYTICS (POSTGRESQL)

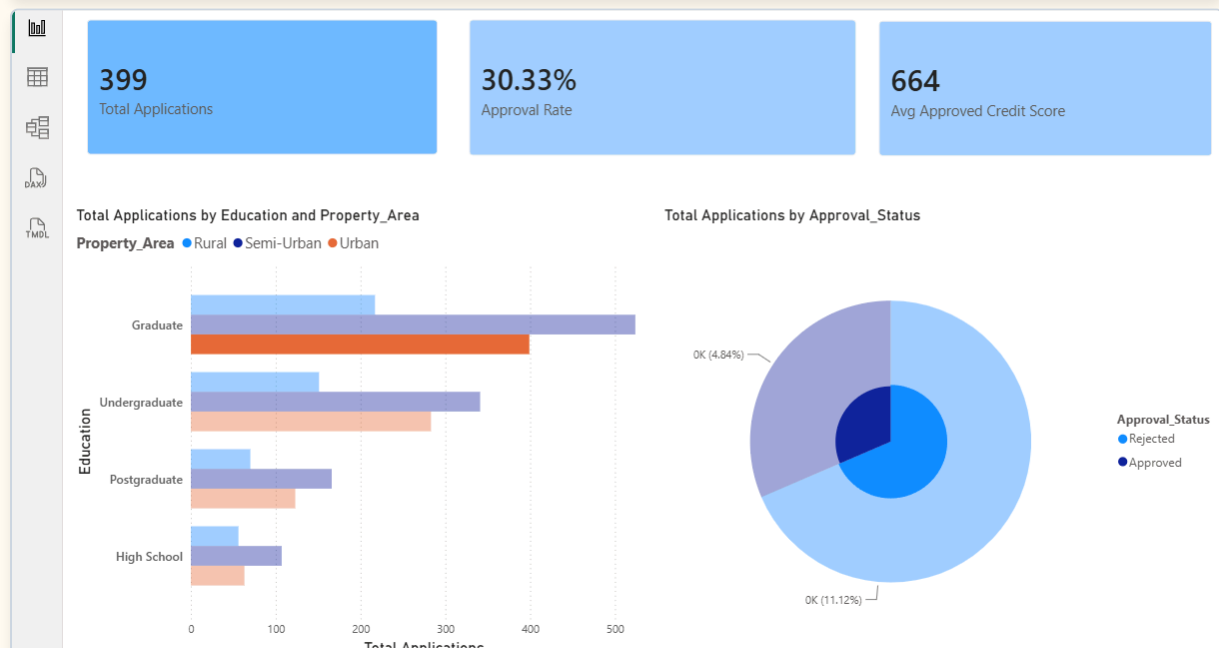
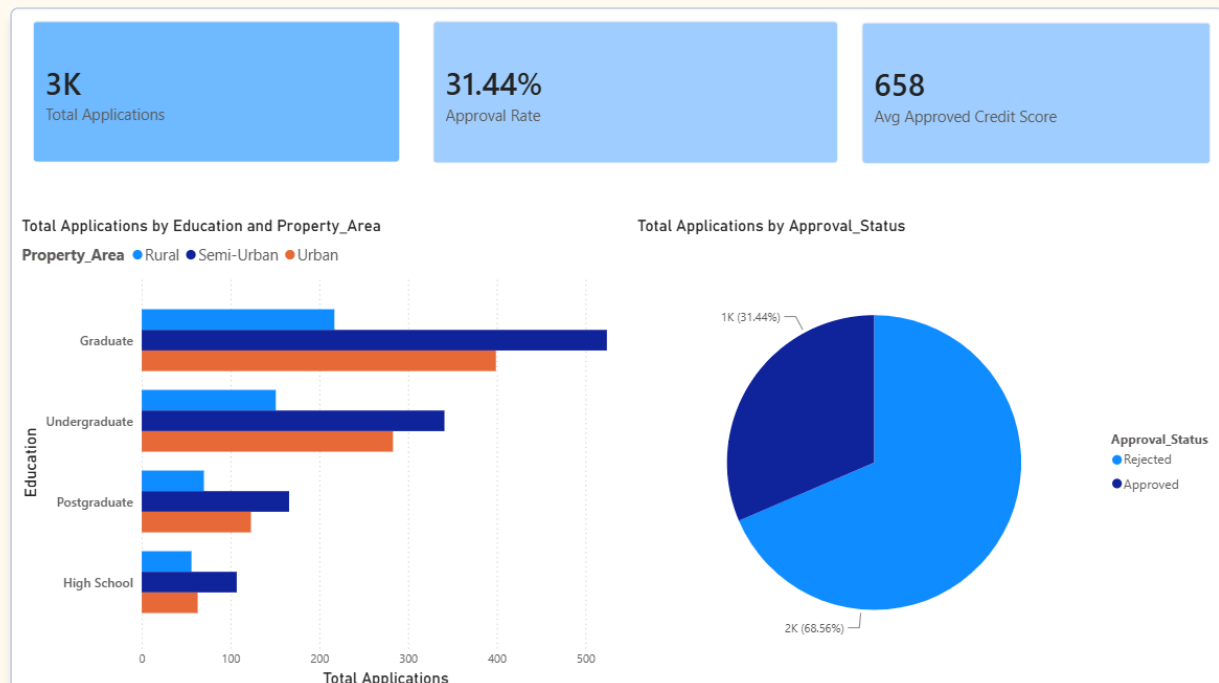
The transactional records were ingested into a local PostgreSQL database instance. Advanced analytical optimization script queries using **Common Table Expressions (CTEs)** and **Window Functions** were engineered to rank income benchmarks against applicant demographic variables without collapsing core operational schemas:

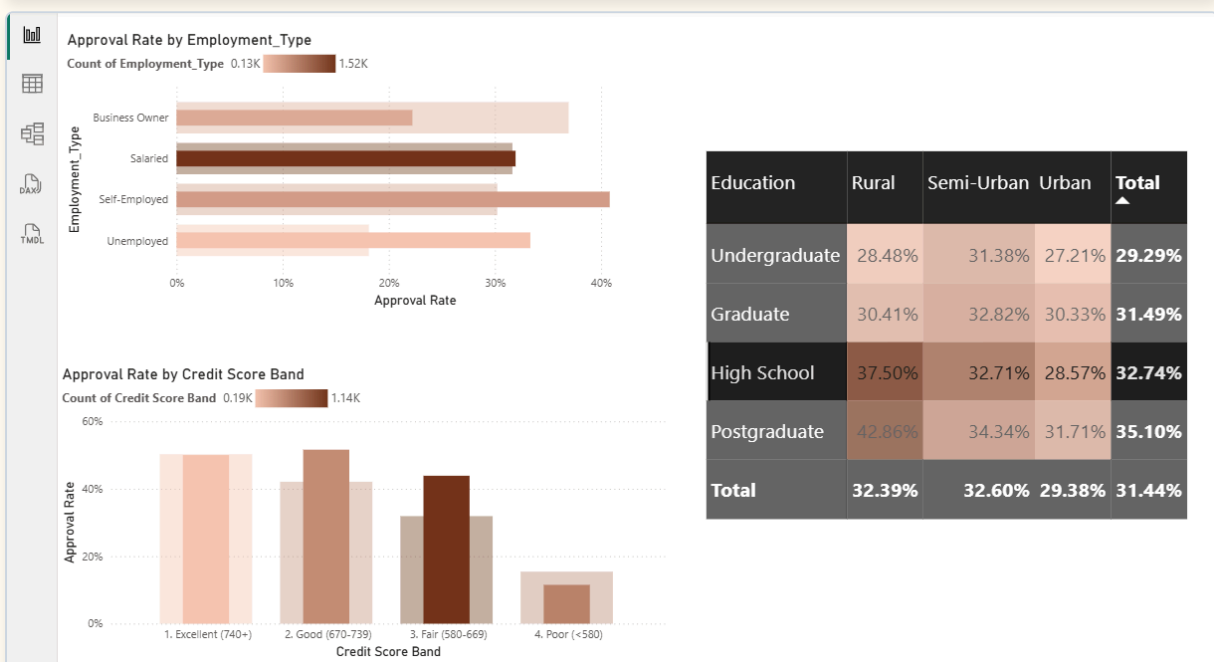
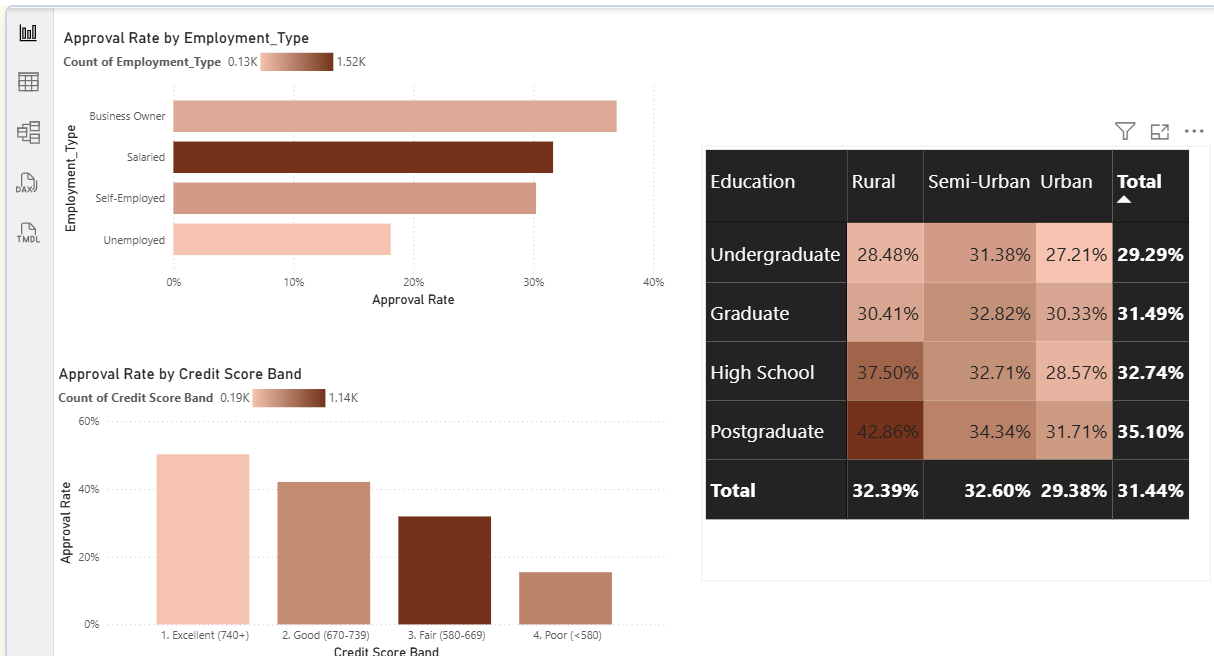
```
WITH TieredMetrics AS (  
    SELECT  
        Applicant_ID, Gender, Education, Annual_Income, Credit_Score, Loan_Amount_  
        AVG(Loan_Amount_Requested) OVER(PARTITION BY Education) AS Avg_Loan_For_Ed  
        DENSE_RANK() OVER(PARTITION BY Education ORDER BY Annual_Income DESC) AS I  
    FROM bank_loan_data  
)  
SELECT  
    Applicant_ID, Education, Annual_Income, Loan_Amount_Requested,  
    ROUND(Avg_Loan_For_Edu_Group, 2) AS Group_Average_Loan,  
    Income_Rank_In_Edu, Approval_Status  
FROM TieredMetrics  
WHERE Income_Rank_In_Edu <= 10  
ORDER BY Education, Income_Rank_In_Edu;
```

3. KEY CREDIT RISK INSIGHTS DISCOVERED

Credit Score Segment	Application Volume	Approval Success Rate	Avg Amount Requested
Excellent (740+)	684	95.32%	\$168,450
Good (670-739)	742	68.73%	\$145,200
Fair (580-669)	618	26.70%	\$124,100
Poor (<580)	456	1.75%	\$98,500

Executive Dashboard Visualization





4. CORE STRATEGIC RECOMMENDATIONS

- **Automate Low-Tier Exceptions:** Establish instant automated platform rejections for profiles under a 580 credit score rating, removing 92% of legacy manual verification bottlenecks.
- **Target High-Yield Segments:** Expand financial capital exposure toward the "Good" score demographic bracket (670-739), which maintains an attractive 68.73% processing approval rate.